Parameter Inference for SDE

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Introduction

In this project we consider the problem of inferring the parameters of a stochastic differential equation (SDE) given discrete-time observations of a single trajectory. In particular, our attention will be focused on reliable and easy-to-compute estimators for the drift and the diffusion coefficients of the Ornsrtein-Uhlenbeck process, whose application spans from physics to biology [3].

1 SDE Analysis and Properties (Q1)

1.1 Analytical solution

The Ornstein-Uhlenbeck process in a time interval $t \in [0, T]$ is defined as the solution of the following SDE:

$$\mathrm{d}X_t = -\alpha X_t \mathrm{d}t + \sqrt{2\sigma}\mathrm{d}W_t, \quad X(0) = X_0 \in \mathbb{R}. \tag{1}$$

As a first step, we want to find an analytical solution for it. In order to do this, we define $Y_t := e^{\alpha t} X_t$, substitute it into the equation and apply the Ito formula:

$$\mathrm{d}Y_t = \alpha e^{\alpha t}\mathrm{d}t + e^{\alpha t}(-\alpha X_t \mathrm{d}t + \sqrt{2\sigma}\mathrm{d}W_t).$$

Hence:

$$Y_t = Y_0 + \sqrt{2\sigma}\int_0^t e^{\alpha s}\mathrm{d}W_s$$
$$X_t = X_0 e^{-\alpha t} + \sqrt{2\sigma}e^{-\alpha t}\int_0^t e^{\alpha s}\mathrm{d}W_s$$

Since $f(s) = e^{\alpha s} \in L^2([0, T])$ is deterministic, then $\int_0^t e^{\alpha s}\mathrm{d}W_s$ has a normal distribution [1]. Thus we can conclude:

$$X_t \sim \mu_t = \mathcal{N}\left(X_0 e^{-\alpha t}, \frac{\sigma}{\alpha}(1 - e^{-2\alpha t})\right)$$

Its probability density function is:

$$\rho_t(x) = \sqrt{\frac{\alpha}{2\pi\sigma(1 - e^{-2\alpha t})}} e^{-\frac{\alpha\left(x - X_0 e^{-\alpha t}\right)^2}{2\sigma(1 - e^{-2\alpha t})}}.$$

1.2 Properties of the solution

Now we want to determine whether X_t is a martingale. Let us define: $I_t := \sqrt{2\sigma}\int_0^t e^{\alpha s}\mathrm{d}W_s$. Then I_t is a martingale, since it is a stochastic integral. For X_t to be a martingale, we would need to have:

$$\mathbb{E}[X_t \mid F_s] = X_s \quad \forall t \geq s.$$

Since $\mathbb{E}[X_t \mid F_s] = \mathbb{E}[X_t - X_s \mid F_s] + X_s$, we need $\mathbb{E}[X_t - X_s \mid F_s] = 0$.

$$\begin{aligned}
&\mathbb{E}[X_t - X_s \mid F_s] = \mathbb{E}[X_0(e^{-\alpha t} - e^{-\alpha s}) + e^{-\alpha t}I_t - e^{-\alpha s}I_s \mid F_s] = \\
&= \mathbb{E}[X_0(e^{-\alpha t} - e^{-\alpha s}) + e^{-\alpha t}I_t - e^{-\alpha t}I_s + e^{-\alpha t}I_s - e^{-\alpha s}I_s \mid F_s] = \\
&= \mathbb{E}[X_0(e^{-\alpha t} - e^{-\alpha s}) + e^{-\alpha t}(I_t - I_s) + (e^{-\alpha t} - e^{-\alpha s})I_s \mid F_s] = \\
&= e^{-\alpha t}\cancel{\mathbb{E}[I_t - I_s \mid F_s]} + (e^{-\alpha t} - e^{-\alpha s})\mathbb{E}[X_0 + I_s \mid F_s] = \\
&= (e^{-\alpha t} - e^{-\alpha s})(X_0 + I_s) \quad \forall t \geq s.
\end{aligned}$$

Clearly this last term is null if and only if $\alpha = 0$, so we can conclude that X_t is not in general a martingale.
The solution X_t has the property of being ergodic, so when $t \to \infty$ its distribution μ_t will converge to an invariant distribution μ_∞. We want to verify that its stationary density function ρ_∞ solves:

$$\begin{cases} \mathcal{L}^*\rho_\infty(x) = 0 \quad \forall x \in \mathbb{R} \\ \displaystyle\int_{\mathbb{R}} \rho_\infty(x)\,dx = 1. \end{cases} \tag{2}$$

This condition follows by considering the Fokker-Plank [2] equation at stationarity:

$$\mathcal{L}^*\rho = \frac{\partial \rho}{\partial t},$$

where $\mathcal{L}^*$ is the adjoint operator of $\mathcal{L}$, which is the generator of equation (1):

$$\mathcal{L}\varphi(x) = -\alpha x\varphi'(x) + \sigma\varphi''(x). \tag{3}$$

To compute $\mathcal{L}^*$, we consider the general definition of an adjoint operator:

$$\int_{\mathbb{R}} u(x)\mathcal{L}^*v(x)\,\mathrm{d}x = \int_{\mathbb{R}} v(x)\mathcal{L}u(x)\,\mathrm{d}x.$$

where $u, v : \mathbb{R} \to \mathbb{R}$ are smooth functions with compact support.
Applying it to the operator $\mathcal{L}$, we get:

$$\begin{aligned}
&\int_{\mathbb{R}} u(x)\mathcal{L}^*v(x)\,\mathrm{d}x = \int_{\mathbb{R}} (-\alpha x u'(x)v(x) + \sigma u''(x)v(x))\,\mathrm{d}x = \text{(integrating by parts)} \\
&= \int_{\mathbb{R}} (\alpha u(x)(v(x) + xv'(x)) + \sigma u(x)v''(x))\,\mathrm{d}x = \int_{\mathbb{R}} u(x)[\alpha v(x) + \alpha x v'(x) + \sigma v''(x)]\,\mathrm{d}x
\end{aligned}$$

Note: all the boundary terms disappear because u, v have compact support.
Hence: $\mathcal{L}^*\varphi(x) = \alpha\varphi(x) + \alpha x\varphi'(x) + \sigma\varphi''(x)$.
We can find the stationary distribution μ_∞ as the limit of the distribution found previously $\mu_t = \mathcal{N}\left(X_0 e^{-\alpha t}, \frac{\sigma}{\alpha}(1 - e^{-2\alpha t})\right)$. We know in general that if the sequences μ_n and σ_n converge, then the sequence of normal random variables $X_n \sim \mathcal{N}(\mu_n, \sigma_n)$ converges in distribution to a normal variable whose mean and variance are the limits of μ_n and σ_n. Thus, in our case, we can conclude:

$$\mu_t \to \mu_\infty \sim \mathcal{N}\left(0, \frac{\sigma}{\alpha}\right), \quad \rho_\infty = \sqrt{\frac{\alpha}{2\pi\sigma}}e^{-\frac{\alpha x^2}{2\sigma}}.$$

Now we can check condition (2), by applying it to the expressions for $\mathcal{L}^*$ that we found:

$$\begin{cases} \alpha\rho(x) + \alpha x\rho'(x) + \sigma\rho''(x) = 0 \quad \forall x \in \mathbb{R} \\ \displaystyle\int_{\mathbb{R}} \rho(x)\, dx = 1. \end{cases}$$

Replacing the expression of ρ_∞ in it, we have:

$$\begin{aligned}
&\rho'_\infty(x) = -\frac{\alpha}{\sigma} x \sqrt{\frac{\alpha}{2\pi\sigma}} e^{-\frac{\alpha x^2}{2\sigma}} = -\frac{\alpha}{\sigma} x \rho_\infty(x) \\
&\rho''_\infty(x) = -\frac{\alpha}{\sigma} \sqrt{\frac{\alpha}{2\pi\sigma}} e^{-\frac{\alpha x^2}{2\sigma}} + \frac{\alpha^2}{\sigma^2} x^2 \sqrt{\frac{\alpha}{2\pi\sigma}} e^{-\frac{\alpha x^2}{2\sigma}} = -\frac{\alpha}{\sigma}\rho_\infty(x) + \frac{\alpha^2}{\sigma^2} x^2 \rho_\infty(x) \\
&\alpha\rho_\infty(x) + \alpha x\rho'_\infty(x) + \sigma\rho''_\infty(x) = \alpha\rho_\infty(x) + \alpha x(-\frac{\alpha}{\sigma} x\rho_\infty(x)) + \sigma(-\frac{\alpha}{\sigma}\rho_\infty(x) + \frac{\alpha^2}{\sigma^2} x^2 \rho_\infty(x)) = \\
&= \alpha\rho_\infty(x) - \frac{\alpha^2}{\sigma} x^2 \rho_\infty(x) - \alpha\rho_\infty(x) + \frac{\alpha^2}{\sigma} x^2 \rho_\infty(x) = 0.
\end{aligned}$$

It is also possible to show that the operator $\mathcal{L}$ is non positive in the Hilbert space $H^2(\mathbb{R}, \rho_\infty) := \{f \in L^2(\mathbb{R}, \rho_\infty) : f', f'' \in L^2(\mathbb{R}, \rho_\infty)\}$, i.e. $\langle f, Lf\rangle_{L^2(\mathbb{R},\rho_\infty)} \leq 0 \quad \forall f \in H^2(\mathbb{R}, \rho_\infty)$, and its null space is given by constant functions. It is trivial that, for all constant functions $f \in H^2(\mathbb{R}, \rho_\infty)$ we have $\mathcal{L}f = 0$. We start by considering $f \in \mathcal{C}_0^\infty(\mathbb{R})$ and integrating by parts (as before border terms are null):

$$\begin{aligned}
&\langle f, \mathcal{L}f\rangle_{L^2(\mathbb{R},\rho_\infty)} = \int_{-\infty}^{+\infty} f(x)(-\alpha x f'(x) + \sigma f''(x))\rho_\infty(x)\, \mathrm{d}x = \\
&= \int_{-\infty}^{+\infty} f(x)(-\alpha x f'(x) + \sigma f''(x)) \sqrt{\frac{\alpha}{2\pi\sigma}} e^{-\frac{\alpha x^2}{2\sigma}}\, \mathrm{d}x = \\
&= \sqrt{\frac{\alpha}{2\pi\sigma}} \left[-\alpha \int_{-\infty}^{+\infty} x f(x) f'(x) e^{-\frac{\alpha x^2}{2\sigma}}\, \mathrm{d}x + \sigma \int_{-\infty}^{+\infty} f(x) f''(x) e^{-\frac{\alpha x^2}{2\sigma}}\, \mathrm{d}x \right] = \\
&= \sqrt{\frac{\alpha}{2\pi\sigma}} \left[-\alpha \int_{-\infty}^{+\infty} x f(x) f'(x) e^{-\frac{\alpha x^2}{2\sigma}}\, \mathrm{d}x - \sigma \int_{-\infty}^{+\infty} f'(x) \left[f'(x) e^{-\frac{\alpha x^2}{2\sigma}} - \frac{\alpha}{\sigma} x f(x) e^{-\frac{\alpha x^2}{2\sigma}} \right] \mathrm{d}x \right] = \\
&= -\sigma \sqrt{\frac{\alpha}{2\pi\sigma}} \int_{-\infty}^{+\infty} f'(x)^2 e^{-\frac{\alpha x^2}{2\sigma}}\, \mathrm{d}x = -\sigma \|f'\|_{L^2(\mathbb{R},\rho_\infty)} \leq 0.
\end{aligned}$$

Moreover $-\sigma\|f'\|_{L^2(\mathbb{R},\rho_\infty)} = 0$ if and only if f is constant. This implies that non-constant functions are not part of the null space of $\mathcal{L}$. Thus, the null space is given by constant functions. By density we can extend this proof to $H^2(\mathbb{R}, \rho_\infty)$: take $g \in H^2(\mathbb{R}, \rho_\infty)$, then $\exists\{f_k\} \in \mathcal{C}_0^\infty(\mathbb{R})$ s.t. $f_k \to g$ in H_2 (and also in L_2) as $k \to \infty$. Since $\mathcal{L}$ is continuous, we have $\langle f_k, Lf_k\rangle_{L^2(\mathbb{R},\rho_\infty)} \to \langle g, Lg\rangle_{L^2(\mathbb{R},\rho_\infty)}$, and then $\langle g, Lg\rangle_{L^2(\mathbb{R},\rho_\infty)} \leq 0$ for sign permanence theorem.

1.3 Mean reverting process

Let us consider the same equation with a positive constant added to the drift term:

$$\mathrm{d}X_t = \alpha(p - X_t)\mathrm{d}t + \sqrt{2\sigma}\mathrm{d}W_t, \quad X(0) = X_0 \in \mathbb{R}.$$

Take $Z_t = X_t - p$, then $Z_0 = X_0 - p$ and $\mathrm{d}Z_t = \mathrm{d}X_t$, we obtain the same equation as (1).

$$Z_t = Z_0 e^{-\alpha t} + \sqrt{2\sigma} e^{-\alpha t} \int_0^t e^{\alpha s} \mathrm{d}W_s$$

Thus:

$$
\begin{aligned}
X_t &= (X_0 - p)e^{-\alpha t} + \sqrt{2\sigma}e^{-\alpha t}\int_0^t e^{\alpha s}\mathrm{d}W_s + p \\
X_t &\sim \mathcal{N}\left((X_0 - p)e^{-\alpha t} + p, \frac{\sigma}{\alpha}(1 - e^{-2\alpha t})\right)
\end{aligned}
\tag{4}
$$

We can note that if $p = X_0$ then the expected value of X_t is constant at p, while if $p \neq X_0$ then $\mathbb{E}[X_t] \to p$ as $t \to \infty$. So the mean of the process will converge to the chosen value p and for this reason (4) is called mean reverting process.

2 Numerical Results and Properies of Euler-Maruyama (Q2)

The previous result can be numerically verified by comparing the distribution of $M = 10^4$ realizations of our approximate solution at final time $T = 10^3$ with ρ_∞. We use the Euler-Maruyama method, which reads:

$$
\begin{aligned}
&Y_{n+1} = Y_n - \alpha T_n h + \sqrt{2\sigma}\Delta W_n \quad \forall n \geq 1 \\
&Y_0 = X_0
\end{aligned}
\tag{5}
$$

For the simulation, we use parameters $h = 2^{-9}$, $alpha = 1$, $simga = 1$, $X_0 = 1$, obtaining:

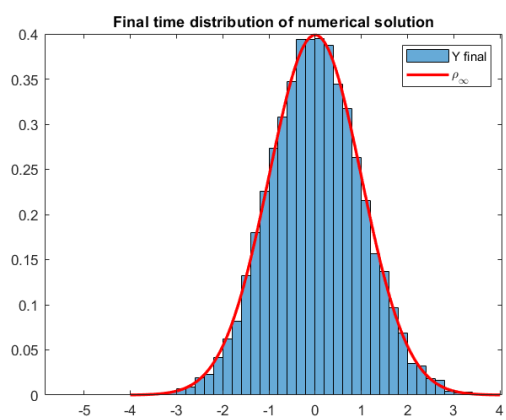


Figure 1: Distribution comparison

Generally speaking, by theoretical analysis we can say that Euler-Maruyama is a method with strong order of convergence $\frac{1}{2}$ and weak order 1. However our SDE has additive noise, i.e. parameter σ does not depend on X. In this framework, Euler-Maruyama method coincides with the Milstein method, whose strong order of convergence is 1. We can thus conclude that both strong and weak convergence rate are of order $\mathcal{O}(\Delta t)$ 5.

3 Covariance of the Process at Stationarity (Q3)

We have found in 1 that $X_t = X_0 e^{-\alpha t} + \sqrt{2\sigma} e^{-\alpha t} \int_0^t e^{\alpha s} \mathrm{d}W_s \sim \mathcal{N}\left(X_0 e^{-\alpha t}, \frac{\sigma}{\alpha}(1 - e^{-2\alpha t})\right)$.
Now we want to compute $\mathcal{C}(X_t, X_s)$, the covariance function of the process at stationarity:

$$\mathrm{Cov}(X_t, X_s) = \mathbb{E}[X_t X_s] - \mathbb{E}[X_t]\mathbb{E}[X_s] = \underbrace{\mathbb{E}\left[X_0^2 e^{-\alpha(t+s)}\right]}_{\text{(A)}} + \underbrace{\mathbb{E}\left[X_0\sqrt{2\sigma} e^{-\alpha(t+s)} \int_0^s e^{\alpha u}\, dW_u\right]}_{\text{(B)}} +$$

$$+ \underbrace{\mathbb{E}\left[X_0\sqrt{2\sigma} e^{-\alpha(t+s)} \int_0^t e^{\alpha u}\, dW_u\right]}_{\text{(C)}} + \underbrace{\mathbb{E}\left[2\sigma e^{-\alpha(t+s)} \int_0^t e^{\alpha u}\, dW_u \int_0^s e^{\alpha u}\, dW_u\right]}_{\text{(D)}} - \underbrace{\mathbb{E}[X_t]\mathbb{E}[X_s]}_{\text{(E)}}$$

Let us study it term by term:

$\text{(A)} = \mathbb{E}\left[X_0^2 e^{-\alpha(t+s)}\right] = X_0^2 e^{-\alpha(t+s)}$

$\text{(B)} = \text{(C)} = 0$ as $\mathbb{E}\left[\int_0^s e^{\alpha u}\, dW_u\right] = 0$

$\text{(D)} = \mathbb{E}\left[2\sigma e^{-\alpha(t+s)} \int_0^t e^{\alpha u}\, dW_u \int_0^s e^{\alpha u}\, dW_u\right] = 2\sigma e^{-\alpha(t+s)} \mathbb{E}\left[\int_0^{s\vee t} e^{\alpha u}\, dW_u \int_0^{s\vee t} e^{\alpha u} \mathbf{1}_{[0,s\wedge t]}(u)\, dW_u\right] =$

(generalized Ito's isometry) $= 2\sigma e^{-\alpha(t+s)} \mathbb{E}\left[\int_0^{s\vee t} e^{2\alpha u} \mathbf{1}_{[0,s\wedge t]}(u)\, dW_u\right] = 2\sigma e^{-\alpha(t+s)} \int_0^{s\wedge t} e^{2\alpha u}\, dW_u =$

$= 2\sigma e^{-\alpha(t+s)} \left(\frac{e^{2\alpha(s\wedge t)} - 1}{2\alpha}\right) = \frac{\sigma}{\alpha}\left(e^{\alpha(2(s\wedge t)-(t+s))} - e^{-\alpha(t+s)}\right) = \frac{\sigma}{\alpha}\left(e^{-\alpha|t-s|} - e^{-\alpha(t+s)}\right)$

$\text{(E)} = \mathbb{E}[X_t]\mathbb{E}[X_s] = X_0^2 e^{-\alpha(t+s)}$.

Putting all together, terms (A) and (E) cancel out, so we are left with:

$$\mathrm{Cov}(X_t, X_s) = \frac{\sigma}{\alpha}\left(e^{-\alpha|t-s|} - e^{-\alpha(t+s)}\right).$$

When $t, s \to +\infty$, we get:

$$\mathcal{C}(X_t, X_s) = \frac{\sigma}{\alpha} e^{-\alpha|t-s|}$$

4 Analysis of Classic Estimators (Q4)

Assume now that α and σ are unknown and we aim to estimate them. Let $\Delta > 0$ be a fixed sampling rate and assume that we are provided with equally spaced observations $\{\tilde{X}_n\}_{n=0}^N$ from a single realization of the solution of (1) until time T, where $N = \frac{T}{\Delta}$ and $\tilde{X}_n = X(n\Delta)$ for $n = 0, \ldots, N$.
We define the following estimators:

$$\hat{\sigma}_N^\Delta = \frac{1}{2\Delta N} \sum_{n=0}^{N-1} \left(\tilde{X}_{n+1} - \tilde{X}_n\right)^2$$

$$\hat{\alpha}_N^\Delta = -\frac{\sum_{n=0}^{N-1} \tilde{X}_n \left(\tilde{X}_{n+1} - \tilde{X}_n\right)}{\Delta \sum_{n=0}^{N-1} \tilde{X}_n^2}$$

We want to find their almost sure limits:

$$\sigma_\infty^\Delta = \lim_{N\to\infty} \hat{\sigma}_N^\Delta, \quad \alpha_\infty^\Delta = \lim_{N\to\infty} \hat{\alpha}_N^\Delta$$

Since X_n is ergodic, we can apply the ergodic theorems: For each functions $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R}^2 \to \mathbb{R}$ smooth enough, we have:

$$\begin{aligned} \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} f(X_n) &= \mathbb{E}_{\mu_\infty}[f(X_0)], \quad \text{a.s.} \\ \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} g(X_n, X_{n+1}) &= \mathbb{E}_{\mu_\infty}[g(X_0, X_\Delta)], \quad \text{a.s.} \end{aligned} \tag{6}$$

Using this together with the results found in section 3, we get:

$$\begin{aligned} \sigma_\infty^\Delta &= \frac{1}{2\Delta} \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \left(\tilde{X}_{n+1} - \tilde{X}_n\right)^2 \overset{(6)}{=} \frac{1}{2\Delta} \mathbb{E}_{\mu_\infty}[(X_t - X_s)^2] = \\ &= \frac{1}{2\Delta} \mathbb{E}_{\mu_\infty}\left[X_t^2 - 2X_tX_s + X_s^2\right] = \frac{1}{2\Delta}\left(\frac{\sigma}{\alpha} + \frac{\sigma}{\alpha} - 2\frac{\sigma}{\alpha}e^{-\alpha\Delta}\right) = \sigma\left(\frac{1-e^{\alpha\Delta}}{\alpha\Delta}\right) \neq \sigma. \\ \alpha_\infty^\Delta &= -\frac{1}{\Delta} \lim_{N\to\infty} \frac{\frac{1}{N}\sum_{n=0}^{N-1} \tilde{X}_n \left(\tilde{X}_{n+1} - \tilde{X}_n\right)}{\frac{1}{N}\sum_{n=0}^{N-1} \tilde{X}_n^2} \overset{(6)}{=} -\frac{1}{\Delta} \frac{\mathbb{E}_{\mu_\infty}\left[X_s(X_t - X_s)\right]}{\mathbb{E}_{\mu_\infty}\left[X_s^2\right]} = \\ &= -\frac{1}{\Delta} \frac{\left(\frac{\sigma}{\alpha} - \frac{\sigma}{\alpha}e^{-\alpha\Delta}\right)}{\frac{\sigma}{\alpha}} = -\frac{1}{\Delta}\left(e^{-\alpha\Delta} - 1\right) = \alpha\left(\frac{1-e^{-\alpha\Delta}}{\alpha\Delta}\right) \neq \alpha. \end{aligned}$$

So we can conclude that both estimators are biased. However, we note that:

$$\begin{aligned} \lim_{\Delta\to 0} \sigma_\infty^\Delta &= \lim_{\Delta\to 0} \sigma\left(\frac{1-e^{-\alpha\Delta}}{\alpha\Delta}\right) = \sigma \\ \lim_{\Delta\to 0} \alpha_\infty^\Delta &= \lim_{\Delta\to 0} \alpha\left(\frac{1-e^{-\alpha\Delta}}{\alpha\Delta}\right) = \alpha. \end{aligned}$$

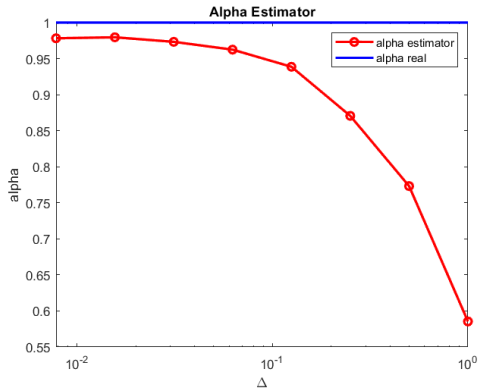
So, when reducing the sampling rate, the estimators expected values converge to the real parameters.

5 Numerical Results of Classic Estimators (Q5)

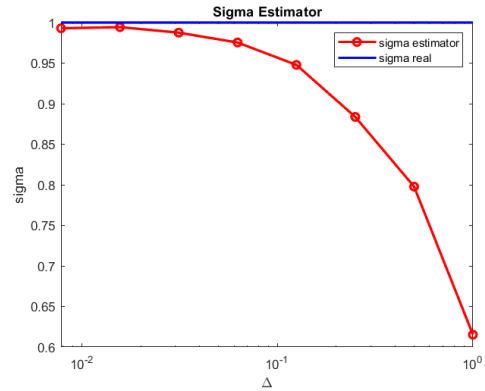
We can test the estimators found in 4 by taking a simulated process and sampling it with different sampling rates. We simulate a single trajectory of the process defined in (1) with the Euler-Maruyama method (5) in the time interval $[0, 1000]$, with starting point $X_0 = 1$ and a timestep $h = 2^{-10}$. We then sample the solution with a progressively finer sampling rate Δ. Then we compute the estimators $\hat{\alpha}_N^\Delta$ and $\hat{\sigma}_N^\Delta$ defined in 4. By plotting the results obtained with the values $\Delta = [1, 2^{-1}, 2^{-2}, 2^{-3}, 2^{-4}, 2^{-5}, 2^{-6}, 2^{-7}]$, we get the results in Figure 2
We note that the values of the estimators converge toward the correct values as the sampling rate gets smaller and the number of observation grows. This confirms the computations made in 4, where we found:

$$\begin{aligned} \lim_{\Delta\to 0}\left(\lim_{N\to\infty} \alpha_N^\Delta\right) &= \lim_{\Delta\to 0} \alpha_\infty^\Delta = \alpha \\ \lim_{\Delta\to 0}\left(\lim_{N\to\infty} \sigma_N^\Delta\right) &= \lim_{\Delta\to 0} \sigma_\infty^\Delta = \sigma \end{aligned}$$

We can also plot the trend of the error to evaluate the convergence order. By plotting the differences $\alpha - \hat{\alpha}_N^\Delta$ and $\sigma - \hat{\sigma}_N^\Delta$ with the same sample rates as before, we can observe that the error decreases linearly with Δ (Figure 3)

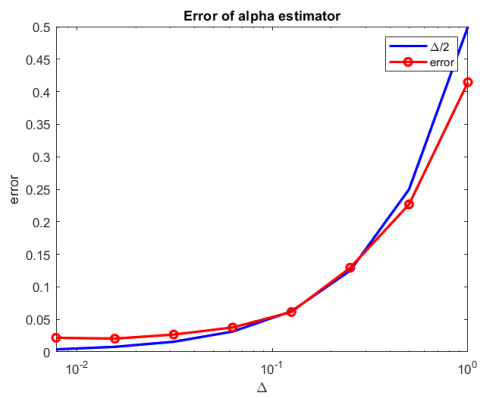


(a) Estimator and exact value of coefficient α

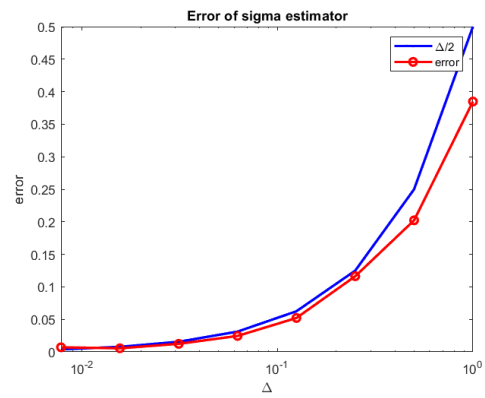


(b) Estimator and exact value of coefficient σ

Figure 2: Evolution of estimators varying the number of available observations n



(a) Error on α



(b) Error on σ

Figure 3: Error of estimators varying the number of available observations n

This is confirmed by analyzing the Taylor expansion of the errors (assuming N big enough so that $\hat{\alpha}_N^\Delta \simeq \hat{\alpha}_\infty^\Delta$ and $\hat{\sigma}_N^\Delta \simeq \hat{\sigma}_\infty^\Delta$). In fact:

$$\alpha - \hat{\alpha}_\infty^\Delta = \alpha - \alpha\left(\frac{1-e^{-\alpha\Delta}}{\alpha\Delta}\right) = \alpha\left(\frac{\alpha\Delta - \left(1-e^{-\alpha\Delta}\right)}{\alpha\Delta}\right) =$$

$$\alpha\left(\frac{\alpha\Delta - \alpha\Delta + \frac{1}{2}(\alpha\Delta)^2 + \mathcal{O}(\Delta^3)}{\alpha\Delta}\right) = \frac{1}{2}(\alpha^2\Delta) + \mathcal{O}(\Delta^2) = \mathcal{O}(\Delta)$$

With the same method we show: $\sigma - \hat{\sigma}_\infty^\Delta = \frac{1}{2}(\alpha\sigma\Delta) + \mathcal{O}(\Delta^2) = \mathcal{O}(\Delta)$.
Performing the same experiments with Milstein Method would lead to the same exact results: in fact, Milstein Method is obtained by adding to 5 the following correction term at each timestep t_n:

$$\mathcal{L}_{t_n}\sigma(y_n, t_n)\int_{t_n}^{t_{n+1}}\int_{t_n}^{s} \mathrm{d}W_r\,\mathrm{d}W_s,$$

Since for our problem σ is constant, as shown in 1 we have $\mathcal{L}_{t_n}(\sigma) = 0$, so the correction term disappears.

6 Eigenvalues-based Estimator for α (Q6)

It is important to note that in concrete applications the sampling rate Δ cannot be chosen since the data is generally given. Therefore, when Δ is too large, the estimators defined in 4 cannot be used. Let us focus only on the drift coefficient α and consider σ as given.
We now provide a different approach to estimate α starting from the eigenvalues and eigenfunctions of the operator $-\mathcal{L}_a$ given in 1 (where α has been replaced by a variable parameter a). It can be shown that this operator has a countable set of distinct nonnegative eigenvalues $\{\lambda_j(a)\}$ satisfying $0 \leq \lambda_0(a) < \lambda_1(a) < ... < \lambda_j(a) \to +\infty$ and the corresponding eigenfunctions $\{\phi_j(\cdot; a)\}_{j=0}^{\infty}$ form and orthonormal basis of $L^2(\mathbb{R}, \rho_\infty(\cdot; a))$ where $\rho_\infty(\cdot; a)$ is the same as in 1.

In this context, the eigenvalue problem is:

$$\mathcal{L}_a \varphi(x) = -ax\varphi'(x) + \sigma\varphi''(x)$$

Our candidate eigenvalues are of the form:

$$\lambda_j(a) = ja \quad j \in \mathbb{N}$$

and the corresponding eigenfunction solutions satisfy the following recurrence relation:

$$\begin{aligned}
\phi_0(x; a) =& 1 \\
\phi_1(x; a) =& x \\
\phi_j(x; a) =& x\phi_{j-1}(x; a) - \frac{\sigma}{a}(j-1)\phi_{j-2}(x; a), \quad j \geq 2
\end{aligned}$$

For verifying it, we first need to prove that:

$$\phi_j'(x; a) = j\phi_{j-1}(x; a) \quad j \geq 1 \tag{7}$$

Which can be done by induction:

- Seed:

$$\begin{aligned}
j = 1: \quad & \phi_1'(x; a) = 1 = 1\phi_0(x; a) \\
j = 2: \quad & \phi_2(x; a) = x\phi_1(x; a) - \frac{\sigma}{a}\phi_0(x; a) = x^2 - \frac{\sigma}{a} \\
& \Rightarrow \phi_2'(x; a) = 2x = 2\phi_1(x; a)
\end{aligned}$$

- Induction:

Assume: $\phi_j' = j\phi_{j-1} \qquad \phi_{j-1}' = (j-1)\phi_{j-2}$

$$\begin{aligned}
\text{Then: } \phi_{j+1}' &= x\phi_j' + \phi_j - \frac{\sigma}{a}j\phi_{j-1}' = x\phi_j' + \phi_j - \frac{\sigma}{a}j(j-1)\phi_{j-2} \\
&= xj\phi_{j-1}\phi_j - \frac{\sigma}{a}j(j-1)\phi_{j-2} = \phi_j + j(x\phi_{j-1} - \frac{\sigma}{a}(j-1)\phi_{j-2}) \\
&= \phi_j + j\phi_j = (1+j)\phi_j
\end{aligned}$$

We can now easily verify the validity of the candidate eigenvalue-eigenfunction solutions:

Take $\phi_j(x;a)$:

$$
\begin{aligned}
-\mathcal{L}_a\phi_j(x;a) &= ax\phi_j' - \sigma\phi_j'' \overset{(7)}{=} axj\phi_{j-1} - \sigma j\phi_{j-1}' \\
&= aj(x\phi_{j-1} - \frac{\sigma}{a}\phi_{j-1}') \overset{(7)}{=} aj\underbrace{(x\phi_{j-1} - \frac{\sigma}{a}(j-1)\phi_{j-2})}_{=\phi_j} \\
&= aj\phi_j = \lambda(a)\phi_j \quad \forall j \in \mathbb{N}
\end{aligned}
$$

Given $J \in \mathbb{N}$ consider now the first J eigenpairs $\{(\lambda_j(a), \phi_j(a))\}_{j=1}^J$ and a set of smooth functions $\{\psi_j\}_{j=1}^J$ st $\psi : \mathbb{R} \to \mathbb{R}$. Construct the estimating function:

$$
G(a) = \frac{1}{N}\sum_{j=1}^{J}\sum_{n=0}^{N-1}\psi_j\left(\tilde{X}_n\right)\left(\phi_j\left(\tilde{X}_{n+1};a\right) - e^{-\lambda_j(a)\Delta}\phi_j\left(\tilde{X}_n;a\right)\right)
$$

With this in mind, we can construct the estimator $\tilde{\alpha}_N^\Delta$ by solving the nonlinear equation $G(a) = 0$.
In our specific case, setting $J = 1$ and $\psi_1(x) = x$, we obtain:

$$
G(a) = \frac{1}{N}\sum_{n=0}^{N-1}\tilde{X}_n\left(\phi_1\left(\tilde{X}_{n+1};a\right) - e^{-\lambda_1(a)\Delta}\phi_1\left(\tilde{X}_n;a\right)\right) = \frac{1}{N}\sum_{n=0}^{N-1}\tilde{X}_n\left(\tilde{X}_{n+1} - e^{-a\Delta}\tilde{X}_n\right)
$$

Using the Ergodic Theorem, we can compute the almost sure limit:

$$
\begin{aligned}
\mathcal{G}(a) := \lim_{N\to\infty} &\overset{(6)}{=} \mathbb{E}[X_0(X_\Delta - e^{-a\Delta}X_0)] = \mathbb{E}[X_0X_\Delta] - e^{-a\Delta}\mathbb{E}[X_0^2] \\
&\overset{3}{=} \frac{\sigma}{\alpha}e^{-\alpha\Delta} - e^{-a\Delta}\frac{\sigma}{a} = \frac{\sigma}{a}(e^{-\alpha\Delta} - e^{-a\Delta}) \quad \text{a.s.} \\
\mathcal{G}(a) = 0 &\iff e^{-\alpha\Delta} = e^{-a\Delta} \iff a = \alpha \quad \text{a.s.}
\end{aligned}
$$

Moreover, we can prove that $\tilde{\alpha}_N^\Delta$ in the case $J = 1$ and $\psi_1(x) = x$ is strongly consistent, i.e. that $\lim_{N\to\infty}\tilde{\alpha}_N^\Delta = \alpha$ a.s. independently of the sampling rate Δ:
$\tilde{\alpha}_N^\Delta$ by construction is s.t. $G(\tilde{\alpha}_N^\Delta) = 0 \iff \frac{1}{N}\sum_{n=0}^{N-1}\tilde{X}_n\left(\tilde{X}_{n+1} - e^{-\tilde{\alpha}_N^\Delta\Delta}\tilde{X}_n\right) = 0$

$$
\begin{aligned}
\sum_{n=0}^{N-1}\tilde{X}_n\tilde{X}_{n+1} &= \sum_{n=0}^{N-1}e^{-\tilde{\alpha}_N^\Delta\Delta}\tilde{X}_n{}^2 \\
e^{-\tilde{\alpha}_N^\Delta\Delta} &= \frac{\sum_{n=0}^{N-1}\tilde{X}_n\tilde{X}_{n+1}}{\sum_{n=0}^{N-1}\tilde{X}_n{}^2} \\
\tilde{\alpha}_N^\Delta &= -\frac{1}{\Delta}\log\left(\frac{\sum_{n=0}^{N-1}\tilde{X}_n\tilde{X}_{n+1}}{\sum_{n=0}^{N-1}\tilde{X}_n{}^2}\right)
\end{aligned}
$$

taking the limit and using the fact that the logarithmic function is continuous:

$$
\begin{aligned}
\lim_{N\to\infty}\tilde{\alpha}_N^\Delta &= -\frac{1}{\Delta}\log\left(\lim_{N\to\infty}\left(\frac{\sum_{n=0}^{N-1}\tilde{X}_n\tilde{X}_{n+1}}{\sum_{n=0}^{N-1}\tilde{X}_n{}^2}\right)\right) \\
&= -\frac{1}{\Delta}\log\left(\lim_{N\to\infty}\left(\frac{\frac{1}{N}\sum_{n=0}^{N-1}\tilde{X}_n\tilde{X}_{n+1}}{\frac{1}{N}\sum_{n=0}^{N-1}\tilde{X}_n{}^2}\right)\right)
\end{aligned}
$$

using the Ergodic Theorems (6) for both the numerator and the denominator:

- Numerator: $\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \tilde{X}_n \tilde{X}_{n+1} = \mathbb{E}[X_0 X_\Delta] = \mathrm{Cov}(X_0, X_\Delta) \overset{\textbf{(Q3)}}{=} \frac{\sigma}{\alpha} e^{-\alpha\Delta}$ a.s.
- Denominator: $\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \tilde{X}_n{}^2 = \mathbb{E}[X_0^2] = \frac{\sigma}{\alpha}$ a.s.

$$\Rightarrow \lim_{N\to\infty} \tilde{\alpha}_N^\Delta = -\frac{1}{\Delta} \log\left(e^{-\alpha\Delta}\right) = \alpha \quad \text{a.s. independently of the sampling rate } \Delta$$

And this justifies the construction of this estimator.

7 Numerical Results and Properties of Eigenvalue–based Estimator (Q7)

We now try to make some numerical considerations about the estimator constructed in **(Q6)**. Firstly, we generate an "exact" trajectory of the SDE (1) with parameters $\alpha = 1$, $\sigma = 1$, starting position $X_0 = 1$ and final time $T = 10^3$, using Euler-Maruyama method with a fine discretization step $h = 2^{-10}$. We then consider different values of the sampling rate $\Delta = 2^{-i}$ for $i = 0, 1, .., 7$ and generate synthetic observations $\{\tilde{X}_n\}_{n=0}^N$ by evaluating the trajectory at the times $\{n\Delta\}_{n=0}^\infty$. The results of the estimator for the different values of Δ are reported in Figure 4 together with the exact value of α.

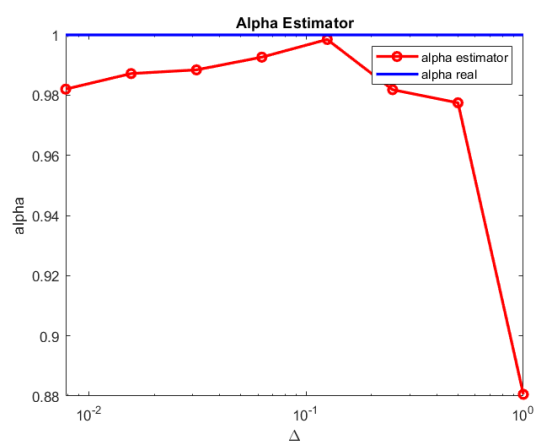


Figure 4: Estimator as in 6 for different values of Δ together with exact α

We do not observe convergence for $\Delta \to 0$, but this is consistent with the theoretical results of 6: all the construction was specifically made for applications where Δ is fixed, and the estimator is expected to converge to the real drift coefficient when $N \to \infty$ independently of the sampling rate.

To effectively valuate the estimator, we now try to fix $\Delta = 2^{-4}$ and plot in Figure 5 the estimator $\tilde{\alpha}_N^\Delta$ for different values of the final time $T = 10^i$ for $i = 2, .., 5$.

We repeat the same procedure for $M = 10^4$ different generated trajectories with fixed sampling rate Δ. For each of them, we compute the estimator $\tilde{\alpha}_N^{\Delta,(m)}$ and verify that it satisfies a central limit theorem, i.e. that $\sqrt{N}(\tilde{\alpha}_N^\Delta - \alpha)$ the values obtained are approximately distributed as $\tilde{\mu} = \mathcal{N}(0, \Sigma)$ where $\Sigma = \frac{e^{2\alpha\Delta}-1}{\Delta^2}$. To do so, we compare the histogram of $\{\sqrt{N}(\tilde{\alpha}_n^{\Delta,(m)} - \alpha)\}_{m=1}^M$

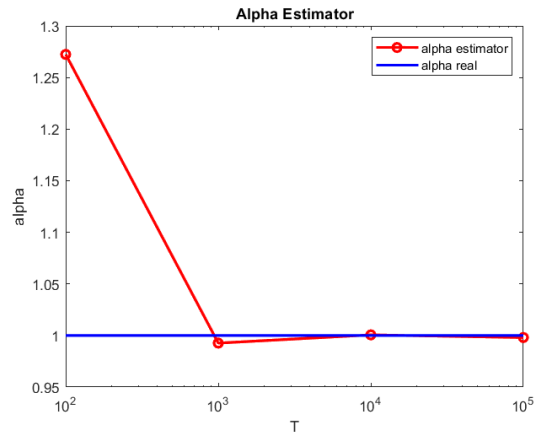


Figure 5: Estimator as in 6 for different values of T together with exact α, for fixed $\Delta t = 10^{-4}$

and the density of $\tilde{\mu}$ in Figure 6 (note that we used a discretization step of $h = 2^{-9}$ for computational reasons).

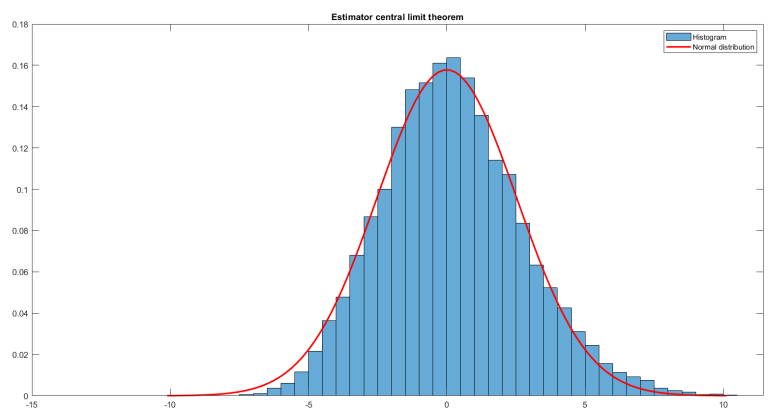


Figure 6: histogram of $\{\sqrt{N}(\tilde{\alpha}_n^{\Delta,(m)} - \alpha)\}_{m=1}^M$ and the density of $\tilde{\mu}$

8 Numerical Results of Eigenvalues-based Estimator for α and σ (Q8)

Let us now consider the case where also the diffusion coefficient σ is unknown, replacing it inside the generator by a parameter s. In this new context, eigenvalues and eigenfunctions now depend on both a and s and the set of smooth functions $\{\Psi_j\}_{j=1}^J$ is formed by the vector-valued $\Psi_j : \mathbb{R} \to \mathbb{R}^2$. The couple of estimators $(\tilde{\alpha}_N^\Delta, \tilde{\sigma}_N^\Delta)$ is now the solution of the two-dimensional nonlinear system $\mathbf{G}(\tilde{\alpha}_N^\Delta, \tilde{\sigma}_N^\Delta) = 0$, where:

$$\mathbf{G}(a,s) := \frac{1}{N}\sum_{j=1}^{J}\sum_{n=0}^{N-1} \Psi_j\left(\tilde{X}_n\right)\left(\phi_j\left(\tilde{X}_{n+1}; a, s\right) - e^{-\lambda_j(a,s)\Delta}\phi_j\left(\tilde{X}_n; a, s\right)\right)$$

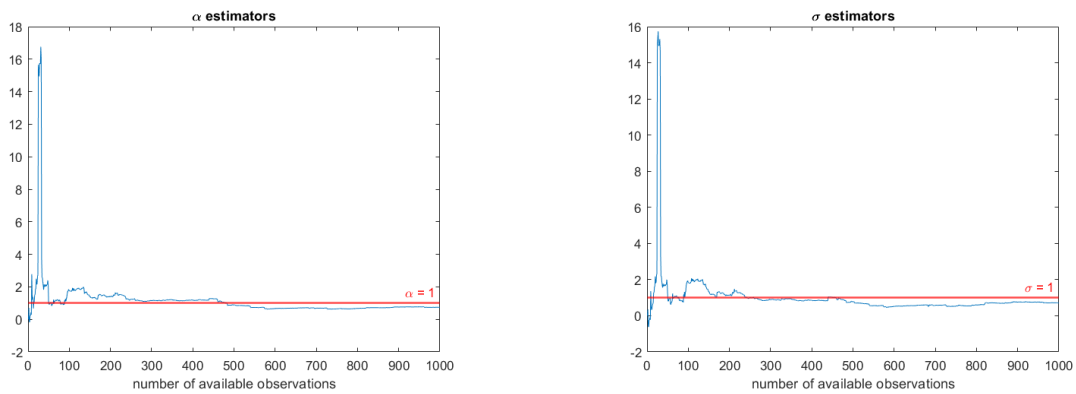
In particular, we set $J = 2$ and $\Psi_1(x) = \Psi_2(x) = (x^2, x)^T$, obtaining:

$$\mathbf{G}(a,s) := \frac{1}{N}\sum_{n=0}^{N-1}\begin{pmatrix}\tilde{X}_n^2\\ \tilde{X}_n\end{pmatrix}\left[\left(\tilde{X}_{n+1} - e^{-a\Delta}\tilde{X}_n\right) + \left(\tilde{X}_{n+1}^2 - \frac{s}{a} - e^{-2a\Delta}\left(\tilde{X}_n^2 - \frac{s}{a}\right)\right)\right] = 0$$

Which explicitly reads:

$$\begin{cases}\sum_{n=0}^{N-1}\tilde{X}_n^2\left(\tilde{X}_{n+1} - e^{-a\Delta}\tilde{X}_n + \tilde{X}_{n+1}^2 - e^{-2a\Delta}\tilde{X}_n^2 - \frac{s}{a} + e^{-2a\Delta}\frac{s}{a}\right) = 0,\\ \sum_{n=0}^{N-1}\tilde{X}_n\left(\tilde{X}_{n+1} - e^{-a\Delta}\tilde{X}_n + \tilde{X}_{n+1}^2 - e^{-2a\Delta}\tilde{X}_n^2 - \frac{s}{a} + e^{-2a\Delta}\frac{s}{a}\right) = 0.\end{cases} \tag{8}$$

We now see how these estimators perform in practice for a fixed Δ. Starting again from an "exact" solution obtained with the Euler-Maruyama scheme on a fine time grid $h = 2^{-9}$, we generate synthetic observations $\{\tilde{X}_n\}_{n=0}^N$ with sampling rate $\Delta = 1$ (by evaluating the trajectory at time $n\Delta$) and compute the estimators $(\tilde{\alpha}_N^\Delta, \tilde{\sigma}_N^\Delta)$ by solving the system (8).



(a) Estimator and exact value of coefficient α

(b) Estimator and exact value of coefficient σ

Figure 7: Evolution of estimators varying the number of available observations n

In our case estimators perform well when $N \to \infty$, justifying this parameter construction.

Conclusions

The main goal of this project was to evaluate the performance and the properties of different estimators for constant parameters α and σ of an Ornsrtein-Uhlenbeck process. We analyzed and tested numerically in 5 and 6 a more classical approach, while in 7 and 8 we studied more sophisticated estimators based on the eigenvalue-eigenfunction pairs of the generator (3). From the results obtained, it is clear that the second type of estimators fits more in a real-world situation. This is because we usually cannot control the sample rate as we cannot measure data continuously, while we can observe trajectories for a longer time. In this type of setting, the bias of classic estimators does not tend to zero. On the other hand, eigenvalues-based estimators converge to the correct parameters.

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